EE4-23
EE9-CS2-1
EE9-FPN2-05

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc and EEE/EIE PART IV: MEng and ACGI

Corrected copy

STABILITY AND CONTROL OF NON-LINEAR SYSTEMS

Tuesday, 7 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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1. Consider the following two-dimensional time-invariant nonlinear system:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_1^3 + x_1 - x_2(1+x_1)^2 \end{aligned}$$

with state $x = [x_1, x_2]'$.

a) Discuss existence and uniqueness of solutions. [2]

b) Find and sketch the nullclines of the vector-field. [3]

c) Identify the regions of the phase-plane in which the nullclines partition $\mathbb{R}^2$ and the orientation of the vector-field in each one of them. [3]

d) Find out, if any of the regions identified in part c) are forward invariant. [2]

e) Find all equilibria of the system. [2]

f) Linearize the system around each equilibrium and discuss the local phase-portrait (discuss hyperbolicity). [3]

g) Assess, for non-hyperbolic equilibria, their local stability or instability properties by using Lyapunov's criteria. [3]

h) Sketch the global phase portrait of the system, emphasizing equilibria and how their local portraits merge into each other. [2]

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2. Consider the nonlinear system given below:

$$\begin{aligned} \dot{x}_1 &= x_2 - x_1 x_2^2 \\ \dot{x}_2 &= -x_1 - x_2 x_1^2, \end{aligned}$$

with state $x \in \mathbb{R}^2$.

a) Analyze the (local) Lyapunov stability property of the equilibrium $x_e = 0$ by choosing a suitable candidate Lyapunov function (Hint: use a quadratic Lyapunov function). [5]

b) Prove global attractivity of the equilibrium (Hint: use Lasalle's invariance principle if needed). [5]

c) Modify the previous system as follows:

$$\begin{aligned} \dot{x}_1 &= x_2 + x_1(1 - x_1^2 - 2x_2^2) \\ \dot{x}_2 &= -x_1, \end{aligned}$$

Compute all equilibrium points and discuss their stability properties. [3]

d) Find a set C that is closed and forward invariant (for the new system's dynamics), which does not contain equilibria. [5]

e) Discuss the type of ω-limit sets expected for initial conditions $x_0 \in C$. [2]

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3. Consider the nonlinear, time-invariant control system of equations:

$$\begin{aligned}\dot{x}_1 &= x_2 + x_3^3 + x_2 u \\ \dot{x}_2 &= -x_1^3 - x_1^3 u \\ \dot{x}_3 &= -x_1^3 + x_3 u\end{aligned}$$

$$y = x_3^4$$

with state $x = [x_1, x_2, x_3]'$ taking values in $\mathbb{R}^3$, input $u \in \mathbb{R}$ and scalar output y.

a) Show that the system is (control) affine. [2]

b) Find a storage function $V(x)$ and show that the system is passive and lossless (Hint: choose a polynomial storage function of the following kind, $V(x) = \alpha_1 x_1^{n_1} + \alpha_2 x_2^{n_2} + \alpha_3 x_3^{n_3}$ for suitable selections of the coefficients α_i and exponents n_i). [7]

c) Design a static output feedback controller, $u = k(y)$, capable of making $x = 0$ a globally asymptotically stable equilibrium (Justify your answer with a detailed stability analysis). [6]

d) Consider the constant input $u(t) = -1$. Show that, for any initial condition, solutions fulfill $x_2(t) = x_2(0)$. [1]

e) Analyze, for $u = 1$, the dynamics of the system on the plane $\{x : x_2 = c\}$ for all constant values $c \in \mathbb{R}$. Compute $\omega(x_0)$ for any $x_0 \in \mathbb{R}^3$. [4]

4. Consider the following nonlinear control system:

$$\begin{aligned}\dot{x}_1 &= x_2\\ \dot{x}_2 &= \operatorname{atan}(x_3)+(2+\sin(x_1+x_2^2))u\\ \dot{x}_3 &= -(x_1+x_3)^3+x_1-x_2\end{aligned}$$

$$y = x_1,$$

with state $x \in \mathbb{R}^3$, scalar input $u \in \mathbb{R}$ and scalar output y.

a) Does the system admit a well-defined relative degree? If so, compute its value and specify whether this is locally or globally defined. [3]

b) Define a feedback $u = k(x, v)$ which linearizes the system from auxiliary input v to output y. [4]

c) Bring the system in normal form and highlight the internal dynamics. (Hint: choose the state of the internal dynamics of the following form $\xi = x_3 + \alpha x_1$, for a suitable $\alpha \in \mathbb{R}$). [4]

d) Discuss the Input-to-State Stability of the internal dynamics. [4]

e) Design a linear stabilizing controller for the chain of integrators resulting from the I/O linearization carried out in b). [3]

f) Discuss whether or not the feedback designed in e) is guaranteed to achieve global asymptotic stability of the whole closed-loop system. [2]